Matteo Papini

Assistant Professor

Contact Information

Work Address	Dipartimento di Informatica "Giovanni degli Antoni", Università degli studi di Milano, via Celoria 18, 20133 Milan, Italy
Email	matteo.papini@unimi.it
Website	`https://t3p.github.io/`
Telephone	+39 02503 16210

Highlights

Matteo Papini is a **Tenure Track Assistant Professor (RTT)** at the **University of Milan**, Italy, in the LAILA lab (Laboratory for Artificial Intelligence and Learning Algorithms) in the Department of Computer Science. His main line of research is **Reinforcement Learning**. His research interests include online learning, multi-armed bandits, learning theory, and multi-agent systems.
In 2021 he earned a PhD in Information Technology (cum laude) from **Politecnico di Milano**, Milan, Italy, with a dissertation on safe policy optimization. In 2020 he worked as a student research intern at **Facebook AI Research** (now Meta). Before he was a postdoctoral researcher in the Artificial Intelligence and Machine Learning Research Group of **Universitat Pompeu Fabra** (UPF), Barcelona, Spain; a fixed-term assistant professor (RTDA) in Politecnico di Milano and a member of the RL3 group (Real Life Reinforcement Learning Research Lab) of AIRLab (Artificial Intelligence and Robotics Laboratory).
He has published more than 20 peer-reviewed research papers in international venues, including top artificial intelligence and machine learning conferences (**NeurIPS**, **ICML**, **COLT**, **AAAI**, **IJCAI**) and journals (**JMLR**, **Machine Learning**).
He is an member of **ELLIS Scholar** since 2026 and a member of ELLIS (European Laboratory for Learning and Intelligent Systems) since 2022. He serves as area chair for NeurIPS (since 2024) and ICML (since 2025) and action editor of **TMLR** (Transactions on Machine Learning Research) since 2024. He co-organized **RLSS** (Reinforcement Learning Summer School) in 2023 (Barcelona) and 2026 (Milan) and was local chair of **ALT 2025** (International Conference on Algorithmic Learning).

Faculty and Research Positions

October 2025–present **Tenure-track Assistant Professor (RTT)**, *Università degli Studi di Milano*, Milan, Italy

2023–2025 **Fixed-term Assistant Professor (RTDA)**, *Politecnico di Milano*, Milan, Italy

2021–2023 **Postdoctoral researcher**, *Universitat Pompeu Fabra*, Barcelona

Mar-Apr 2021 **Research Assistant** (Attività di Supporto alla Ricerca), *Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano*, Milan, Italy
Research Topic: Reinforcement Learning Techniques for Developing Artificial Test Drivers on a F1 Simulator, Principal Investigator: Marcello Restelli

Sep-Dec 2020 **Research Intern at Facebook AI Research**, working (remotely) with the Paris FAIR team, under the supervision of Matteo Pirotta. Research topic: representation learning for contextual bandits. The project led to the publication of a paper at ICML 2021 [C3]

2017–2021 **Research Assistant** (Assegno di Ricerca), *Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano*, Milan, Italy
Research topic: *Study and Development of Reinforcement Learning Techniques in Industrial Settings*, Principal Investigator: Marcello Restelli.

Education

2017–2021 **Ph.D., Ingegneria dell'Informazione / Information Technology**, Politecnico di Milano, Milan, Italy
Advisor: Prof. Marcello Restelli
Dissertation: *Safe Policy Optimization* (`http://hdl.handle.net/10589/170196`)
Awarded cum laude
Date of degree: March 11, 2021

2015–2017 **M.Sc., Computer Science and Engineering - Ingegneria Informatica**, *Politecnico di Milano, Milan, Italy*
Master Thesis: *Adaptive Batch Size for Safe Policy Gradient Methods*, supervised by Prof. Marcello Restelli
Final mark: 110/110 cum laude
Date of degree: July 27, 2017

2012–2015 **B.Sc., Ingegneria Informatica (Computer Engineering)**, *Politecnico di Milano, Milan, Italy*
Final mark: 110/110 cum laude
Date of degree: July 24, 2015

2007–2012 **High School Diploma (Liceo Scientifico PNI)**, *Ist. Istr. Sup. P.L. Nervi*, Morbegno (SO), Italy, Final mark: 100/100

Author Profile and Metrics (as of October 14, 2026)

Google Scholar **h-index 17**, `https://scholar.google.it/citations?user=A2WxZlsAAAAJ`

Scopus **h-index 13**, *`https://www.scopus.com/authid/detail.uri?authorId=57202057824`*

ORCiD https://orcid.org/0000-0002-3807-3171

dblp https://dblp.uni-trier.de/pid/209/4897.html

Summary of Publications

Journal/Conference	# Papers	Scimago 24/CORE 23	Homepage
JMLR	1	Q1	http://www.jmlr.org/
Machine Learning	3	Q1	https://www.springer.com/journal/10994
NeurIPS	7	A*	https://nips.cc/
ICML	8	A*	https://icml.cc/
AAAI	3	A*	https://aaai.org/conference/aaai/
IJCAI	2	A*	https://www.ijcai.org/
COLT	2	A*	https://learningtheory.org/
AISTATS	2	A	https://aistats.org/
AAMAS	1	A	https://www.ifaamas.org/
ALT	2	B	http://algorithmiclearningtheory.org/
IJCNN	1	B	https://ijcnn.org/

Selected Publications

[1] **M. Papini**, D. Binaghi, G. Canonaco, M. Pirotta, M. Restelli: *Stochastic variance-reduced policy gradient.* **ICML** (2018) — **CORE A***

[2] A.M. Metelli, **M. Papini**, F. Faccio, M. Restelli: *Policy optimization via importance sampling.* **NeurIPS (oral, top 3%)** (2018) — **CORE A***

[3] L. Bisi, L. Sabbioni, E. Vittori, **M. Papini**, M. Restelli: *Risk-averse trust region optimization for reward-volatility reduction.* **IJCAI** (2020) — **CORE A***

[4] A.M. Metelli, **M. Papini**, N. Montali, M. Restelli: *Importance Sampling Techniques for Policy Optimization.* **Journal of Machine Learning Research (JMLR)** 21.141 (2020) — **Scimago Q1**

[5] **M. Papini**, M. Pirotta, M. Restelli: *Adaptive batch size for safe policy gradients.* **NeurIPS** (2017) — **CORE A***

[6] P. D'Oro, A. M. Metelli, A. Tirinzoni, **M. Papini**, M. Restelli: *Gradient-aware model-based policy search.* **AAAI** (2020) — **CORE A***

[7] **M. Papini**, A.M. Metelli, L. Lupo, M. Restelli: *Optimistic policy optimization via multiple importance sampling.* **ICML** (2019) — **CORE A***

[8] **M. Papini**, M. Pirotta, M. Restelli: *Smoothing policies and safe policy gradients.* **Machine Learning** (2022) — **Scimago Q1**

[9] **M. Papini**, A. Tirinzoni, M. Restelli, A. Lazaric, M. Pirotta: *Leveraging good representations in linear contextual bandits.* **ICML** (2021) — **CORE A***

[10] **M. Papini**, A. Tirinzoni, A. Pacchiano, M. Restelli, A. Lazaric, M. Pirotta: *Reinforcement learning in linear mdps: Constant regret and representation selection.* **NeurIPS** (2021) — **CORE A***

[11] G. Neu, J. Olkhovskaya, **M. Papini**, L. Schwartz: *Lifting the Information Ratio: An Information-Theoretic Analysis of Thompson Sampling for Contextual Bandits.* **NeurIPS** (2022) — **CORE A*** (authors in alphabetical order)

[12] G. Paczolay., **M. Papini**, A. M. Metelli, I. Harmati, M. Restelli: *Sample complexity of variance-reduced policy gradient: weaker assumptions and lower bounds.* **Machine Learning** (2024) — **Scimago Q1**

Awards and other Recognitions

Oct 2026 **ELLIS Scholar**, *https://ellis.eu/*

Nov 16, 2022 **Recipient of Ayudas Juan de la Cierva-Formación 2021**, *research grant for postdoctoral researchers instructed by the Spanish research agency (Agencia Estatal de Investigación), funded by NextGenerationEU*
Role: researcher (Investigador). Ranked third in national call for the area of Information and Communication Technologies.
https://www.aei.gob.es/convocatorias/buscador-convocatorias/ayudas-contratos-juan-cierva-formacion-2021

Feb 2022 **Member of ELLIS** , *https://ellis.eu/*

2021 **Neurips outstanding reviewer award**, *top 8%*

2020 **ICML top reviewer** certificate of appreciation, *top 33%*

2019 **ICML travel award**, *$1300 USD*

2018 **Oral presentation at NeurIPS**, *top 3%*

2018 **Neurips travel award**, *$1000 USD*

2018 **ICML travel award**, *$1500 USD*

2017 **Neurips travel award**, *$1200 USD*

Participation to Research Projects

2025-present **Team Member**, *European Lighthouse of AI for Sustainability (ELIAS)*
Horizon Europe Global Challenge-RIA/IA/CSA
Principal Investigator: Nicolò Cesa-Bianchi

2024-2025 **Principal Investigator**, *Sviluppo di strategie automatiche di trading per il mercato dell'energia basate su Reinforcement Learning (development of automatic trading strategies for the energy market based on Reinforcement Learning)*, research contract between Politecnico di Milano and A2A S.p.A.

2024-2025 **Principal Investigator**, *CC Auto Tune: ottimizzazione automatica dei parametri del Performance Controller per Compressori Centrifughi (automatic optimization of performance controller parameters for centrifugal compressors)*, research contract between Politecnico di Milano and MADE S.C.A.R.L. (Competence Center for Industry 4.0)
Industrial research project with Baker Hughes Nuovo Pignone.

2023-present **Research Scientist**, *Artificial Intelligence Foundations for Sequential Decision Making*
Extended Partnership - Future Artificial Intelligence Research (FAIR). National Recovery and Resilience Plan, Mission 4 "Education and research" - Component 2 "From research to business" - Investment 1.3, funded by the European Union - NextGenerationEU
Principal Investigator: Nicola Gatti

2021-2022 **Research Scientist**, *Provably Efficient Algorithms for Large-Scale Reinforcement Learning*
ERC (Grant agreement No. 950180)
Principal Investigator: Gergely Neu

2021 **Research Scientist**, *Reinforcement learning techniques for developing artificial test drivers on a F1 simulator*, Ferrari
Industrial project.
Principal Investigator: Marcello Restelli

2017-2019 **Research Scientist**, *TOTAL EFFICIENCY 4.0, (POR FESR)*
In collaboration with Pirelli Tyre S.p.A.
Principal Investigator: Marcello Restelli

Editorial Activities

2024-present **Action Editor** for Transactions on Machine Learning Research (TMLR)

2024-present **Area Chair** for NeurIPS

2025-present **Area Chair** for ICML, RLC (Reinforcement Learning Conference)

2026 Reviewer for COLT

2025 Reviewer for AISTATS

2024-present **Expert reviewer** for Transactions on Machine Learning Research (TMLR)

2024 Reviewer for ICML, COLT, Senior Reviewer for RLC (Reinforcement Learning Conference)

2022-2024 Reviewer for Transactions on Machine Learning Research (TMLR)

2023 Reviewer for ICML, COLT, EWRL, NeurIPS, ICLR, AISTATS

2022 Reviewer for IEEE Transactions on Automatic Control (TACON)

2022 Reviewer for ICML, NeurIPS, EWRL

2021 **Outstanding Reviewer** (top 8%) for NeurIPS, **expert reviewer** for ICML, reviewer for AISTATS, emergency reviewer for AAAI

2020 **Top 33% reviewer** for ICML, reviewer for NeurIPS, AISTATS, AAAI, UAI, ECAI

2019 Reviewer for ICML, NeurIPS, UAI

Organization of Scientific Events

2026 **Reinforcement Learning Summer School (RLSS 2026)**, *June 3–12 2026, Politecnico di Milano, Milan, Italy. Organized with Gianmarco Genalti (Polimi), Alberto Maria Metelli (Polimi), Emmanuel Esposito (Unimi)*, `https://2026.rlsummerschool.com/`

2025 **Interactive Learning and Interventional Representations (ILIR) Ellis UnConference workshop**, *December 2, 2026, Bella Center, Copenhagen, Denmark. co-located with EurIPS. Organized with Nicolò Cesa-Bianchi (Unimi), Andreas Krause (ETH), Bernhard Schölkopf (MPI-IS)* , `https://sites.google.com/view/ilir-workshop-dec-2-2025/`

2025 **The 36th International Conference on Algorithmic Learning Theory (ALT 2025)**, *February 24-27 2025, Milan, Italy. Local Chair, with Giulia Clerici (Ellis Unit Milan)*, `http://algorithmiclearningtheory.org/alt2025/`

2023 **Reinforcement Learning Summer School (RLSS 2023)**, *June 28–July 5 2023, Universitat Pompeu Fabra, Barcelona, Spain. Organized with Vincent Adam (UPF)*, `https://rlsummerschool.com/2023/`

Invited and Contributed Talks

Oct 9, 2026 **Invited talk at Reinforcement Learning and Game Theory workshop**, *AI-ML Systems 2026, Politecnico di Milano, Lecco, Italy, "Adversarial Combinatorial Bandits with Winner Feedback"*

Jul 23, 2026 **Seminar at RIKEN AIP**, *Tokyo, Japan, "Reusing Data in Policy Gradients to Improve Sample Efficiency", hosted by prof. Takayuki Osa*

May 8, 2026 **Short talk at ELLIS Milan + RIKEN AIP Joint ELIAS Workshop**, *University of Milan, Italy, "Laplacian Representations for Reinforcement Learning"*

Sep 18, 2025 **Contributed talk at EWRL 2025**, *(European Workshop on Reinforcement Learning) Tübingen, Germany, "Accelerating Policy Gradient Algorithms with Data Reuse"*

Jan 12, 2024 **Invited talk at the "Mini-Workshop on Reinforcement Learning"**, *University of Mannheim, Germany*

May 25, 2023 **Invited talk at the Theory of Reinforcement Learning Workshop**, *University of Alberta, Edmonton, Canada*, titled "Offline Primal-Dual Reinforcement Learning for Linear MDPs"

May 11, 2023 **Invited talk at the AI Seminars**, *Politecnico di Milano, Milan, Italy*, titled "Large-Scale Offline Reinforcement Learning"

Sep 16, 2022 **Invited talk at the Reinforcement Learning Conference**, *Technische Universität Dresden, Dresden, Germany*, titled "Lifting the Information Ratio"

Jul 1, 2021 **Invited talk at Mathematical Statistics and Learning**, *Barcelona Graduate School of Economics, Barcelona, Spain*, titled "Leveraging Good Representations in Linear Contextual Bandits and MDPs"

Nov 6, 2020 **Invited talk at Gerhard Neumann's research seminar**, *Karlsruhe Institute of Technology (delivered online)*, titled "Safe Policy Optimization"

Sep 19, 2019 **Invited talk at the Workshop on Markets, Algorithms, Prediction and LEarning (MAPLE)**, *Politecnico di Milano, Milan, Italy*, titled "Optimistic Policy Optimization via Multiple Importance Sampling"

Student Supervision and Mentoring

As PhD co-Advisor

2025-present **Leonardo Cesani**, Information Technology, Politecnico di Milano
Co-supervisor with Marcello Restelli

2024-present **Andrea Menta**, Information Technology, Politecnico di Milano
Co-supervisor with Marcello Restelli

2023-present **Alessandro Montenegro**, Information Technology, Politecnico di Milano
Co-supervisor with Alberto Maria Metelli

As Master Thesis Advisor

2023-2026 **Master theses**, *Supervising 2 master students at University of Milan for M.Sc. Computer science, supervised 4 master students for M.Sc. Computer Science and Engineering*, one for M.Sc. Mathematical Engineering, Politecnico di Milano
Relatore of 5 deposited M.Sc. theses (`https://www.politesi.polimi.it/`)

As Master Thesis co-Advisor

2018-2020 and 2023-2025 **Master theses**, *co-supervised several Master theses (M.Sc. Computer Science and Engineering, Mathematical Engineering, Management Engineering)*, Politecnico di Milano
Correlatore of 9 deposited M.Sc. theses (`https://www.politesi.polimi.it/`)

Other Mentoring Activities

2026 **Mentor**, *Women in Machine Learning Symposium, ICML 2026, Seoul, South Korea*

Teaching

2026 **Lecturer**, *Università degli Studi di Milano, Milan, Italy*
Reinforcement Learning, M.Sc. Data Science for Economics and Health, 20 hours, II semester 2026/2027 (scheduled)

2026 **Lecturer**, *Università degli Studi di Milano, Milan, Italy*
Machine Learning, Artificial Neural Networks and Deep Learning (Module 1), B.Sc. Artificial Intelligence, 32 hours + 12 lab hours, I semester 2026/2027 (current)

2026 **Lecturer**, *Università degli Studi di Milano, Milan, Italy*
Reinforcement Learning, M.Sc. Data Science for Economics and Health, 10 hours, II semester 2025/2026

2025 **Lecturer**, *Università degli Studi di Milano, Milan, Italy*
Computer science for cultural mediators (Dati, web e contenuti digitali), B.Sc. Mediazione Linguistica e Culturale, 30 hours, I semester 2025/2026

2025 **Lecturer**, *Politecnico di Milano, Milan, Italy*
Algorithms and data structures (Algoritmi e Principi dell'Informatica – Mod. 2), B.Sc. Computer Science and Engineering, 30 hours, II semester 2024/2025

2025 **Lecturer**, *Politecnico di Milano, Milan, Italy*
Advanced Deep Learning, PhD Information Technology, 10 hours, 2024/2025

2024 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Foundations of Artificial Intelligence, resp. Prof. Francesco Amigoni and Pierluca Lanzi, 16 hours, I semester 2024/2025

2024 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Informatica (computer science for civil engineering students), resp. Prof. Marcello Restelli, 24 hours, II semester 2023/2024

2024 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Machine Learning, resp. Prof. Marcello Restelli, 20 hours, II semester 2023/2024

2023 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Foundations of Artificial Intelligence, resp. Prof. Francesco Amigoni and Pierluca Lanzi, 16 hours, I semester 2023/2024

11-15 Jul 2022 **Teaching Assistant**, *Reinforcement Learning Summer School, Vrije Universitet Amsterdam, Amsterdam, the Netherlands*
Role: delivered a tutorial on implementing policy-gradient algorithms with JAX, tutored students in practical sessions.

2022 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Intelligenza Artificiale (Artificial Intelligence — online class), resp. Prof. Andrea Bonarini, 6 hours, attività di didattica integrativa, II semester 2021/2022

2021 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Intelligenza Artificiale (Artificial Intelligence — online class), resp. Prof. Andrea Bonarini, 6 hours, attività di didattica integrativa, II semester 2020/2021

2020 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Intelligenza Artificiale (Artificial Intelligence — online class), resp. Prof. Andrea Bonarini, 6 hours, attività di didattica integrativa, II semester 2019/2020

2019 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Informatica B (computer science for mechanical engineering students), resp. Prof. Luca Cassano, 26 hours, attività di didattica integrativa (esercitatori), I semester 2019/2020

2018 **Teaching Assistant**, *Politecnico di Milano, Milan, Italy*
Informatica B (computer science for mechanical engineering students), resp. Prof. Luca Cassano, 28 hours, attività di didattica integrativa (esercitatori), I semester 2018/2019

1-12 Jul 2019 **Teaching Assistant**, *Reinforcement Learning Summer School, Inria Lille-Nord Europe, Lille, France*, Role: tutored students in practical sessions

2018 **Teaching Assistant**, *Politecnico di Milano (Polo Territoriale di Como), Como, Italy*
Web and Internet Economics, resp. Prof. Nicola Gatti, 10 hours, *attività di didattica integrativa (esercitatori), II semester 2017-2018*

2017 **Lab Assistant**, *Politecnico di Milano, Milan, Italy*
Informatica B (computer science for mechanical engineering students), resp. Prof. Luca Cassano, 9 hours, attività di didattica integrativa sperimentale (ex responsabili di laboratorio), I semester 2017-2018

2016 **Lab Tutor**, *Politecnico di Milano, Milan, Italy*
Prova Finale–Ingegneria del Software (software engineering: final project), resp. Prof. Carlo Ghezzi, 32 hours, *attività di tutorato (ex tutor di laboratorio), II semester 2015/2016*

Participation on Committees

Nov 16 2026 **PhD Grading Committee Member**, *KTH, Stockholm (scheduled)*

2025-present **Committee member for degree examinations**, *B. Sc. and M. Sc. in Computer Science, University of Milan (several instances)*

April 16 2025 **PhD Jury Member**, *University of Liège, Belgium*

April 10 2025 **President of committee for PhD defense**, *PhD program in Information Technology, Politecnico di Milano*

2024 **Examiner of PhD thesis**, *University of Melbourne, Australia*

June 27 2024 **President of committee for PhD defense**, *PhD program in Information Technology, Politecnico di Milano*

May 2024 **Committee member for assignment of industrial research funding**, *Bando a cascata del progetto "Future Artificial Intelligence Research – FAIR", (Bando Imprese), codice PE0000013, PNRR, Missione 4, Componente 2, Investimento 1.3*

2023-2025 **Committee member for degree examinations**, *B. Sc. and M. Sc. in Computer Science and Enginnering, Politecnico di Milano (several instances)*

Mar 30 2023 **Board member for the Thesis Proposal Defenses** , *PhD program in Information and Communication Technologies, Universitat Pompeu Fabra, Barcelona, Spain*

2023 **Jury Member for the CLAIRE R2Net 2022 Papers Highlights**

2021, 2022 **Evaluator** for the pre-screening of applicants to the ELLIS PhD program

July 5 2022 **Jury member for the undergraduate thesis defenses** , *Universitat Pompeu Fabra, Barcelona, Spain*

Mar 17 2022 **Board member for the Thesis Proposal Defense** , *PhD program in Information and Communication Technologies, Universitat Pompeu Fabra, Barcelona, Spain*

Attendance to Conferences and Workshops

2026 **ICML**, *Seoul, South Korea*, 2 posters + 1 at the CoLorAI workshop + 1 at the NExT-Game workshop

2025 **EurIPS + ELLIS UnConference**, *Copenhagen, Denmark*, organized 1 ELLIS UnConference workshop, 1 poster at ELLIS UnConference, 1 poster at Salon de Refusés

2025 **EWRL 18**, *Tübingen, Germany*, 1 contributed talk, 1 poster

2025 **RLC (Reinforcement Learning Conference)**, *Edmonton, Canada*, 1 poster + 1 spotlight talk at the Workshop on Programmatic Reinforcement Learning

2024 **NeurIPS**, *Vancouver, Canada*, 2 posters

2024 **ICML**, *Vienna, Austria*, 2 posters + 1 at the ARLET workshop

2024 **AISTATS**, *Valencia, Spain*, 1 poster

2024 **ALT**, *San Diego, USA*, 1 poster

2024 **Mini Workshop on Reinforcement Learning**, *Mannheim, Germany*, 1 invited talk

2023 **EWRL 16**, *Brussels, Belgium*, 2 posters

2023 **AISTATS**, *Valencia, Spain*

2023 **Upper Bound**, *Edmonton, Canada*, invited talk at the RL Theory Workshop

2023 **ALT**, *Singapore*, attended as co-author and served as **session chair**

2022 **NeurIPS**, *New Orleans, USA*, presented 2 posters in poster sessions

2022 **ELLIS ILIR Workshop**, *Feldberg, Germany*

2022 **EWRL 15**, *Milan, Italy*, presented 2 posters in poster sessions

2022 **Reinforcement Learning Conference**, *Dresden, Germany*, invited talk

2022 **ICML-2022 Workshop on Complex Feedback in Online Learning**, *Baltimore, USA*, presented 1 poster in poster session

2021 **NeurIPS**, *online edition*, presented 1 poster in online poster session and recorded 1 spotlight talk

2021 **Mathematical Statistics and Learning**, *Barcelona, Spain*, invited talk

2021 **ICML**, *online edition*, presented 1 poster in online poster session and recorded 1 talk

2021 **AAAI**, *online edition*, presented 1 poster in online poster session

2021 **IJCAI**, *online edition*, presented 1 poster in online poster session

2020 **NeurIPS**, *online edition*, attended online

2020 **AISTATS**, *online edition*, recorded 1 talk

2019 **Workshop MAPLE**, *Milan, Italy*, invited talk

2019 **ICML**, *Long Beach, USA*, presented 1 poster and delivered 1 oral presentation

2018 **NeurIPS**, *Montreal, Canada*, presented 1 poster and delivered 1 oral presentation

2018 **EWRL 14**, *Lille, France*, presented 1 poster

2018 **ICML**, *Stockholm, Sweden*, presented 1 poster and delivered 1 oral presentation

2017 **NeurIPS**, *Long Beach, USA*, presented 1 poster

Complete List of Papers

Journal Papers

[J1] L. Civitavecchia, M. Papini. "Exploration-Free Reinforcement Learning with Linear Function Approximation." Reinforcement Learning Journal, vol. 6, 2025, pp. 1856–1879. `https://rlj.cs.umass.edu/2025/papers/Paper194.html`

[J2] G. Tedeschi, M. Papini, A. M. Metelli, and M. Restelli. Search or Split: Policy Gradient with Adaptive Policy Space. Machine Learning 114, 186 (2025). `https://doi.org/10.1007/s10994-025-06820-2`

[J3] G. Paczolay, M. Papini, A. M. Metelli, I. Harmati, and M. Restelli. Sample Complexity of Variance-Reduced Policy Gradient: Weaker Assumptions and Lower Bounds. Machine Learning 113, 6475–6510 (2024). `https://doi.org/10.1007/s10994-024-06573-4`

[J4] M. Papini, G. Manganini, A. M. Metelli, and M. Restelli. "Policy Gradient with Active Importance Sampling." Reinforcement Learning Journal, vol. 2, 2024, pp. 645–675. `https://rlj.cs.umass.edu/2024/papers/Paper90.html`

[J5] M. Papini, M. Pirotta, and M. Restelli. Smoothing policies and safe policy gradients. Machine Learning 111, 4081–4137 (2022). `https://doi.org/10.1007/s10994-022-06232-6`

[J6] A. M. Metelli, M. Papini, N. Montali, M. Restelli. Importance Sampling Techniques for Policy Optimization. Journal of Machine Learning Research 21(141):1–75, 2020. `https://www.jmlr.org/papers/v21/20-124.html`

Conference Papers

[C1] P. Olivieri, F. Lasca, A. Gianola, and M. Papini. Do it for HER: first-order temporal logic reward specification in reinforcement learning. In *AAAI*, pages 24613–24621. AAAI Press, 2026.

[C2] A. Montenegro, F. Mansutti, M. Mussi, M. Papini, and A. M. Metelli. Reusing trajectories in policy gradients enables fast convergence. *ICML (to appear)*, 2026.

[C3] S. Livella, L. Bolis, S. Patania, M. Papini, and D. Ognibene. Mitigating problematic social media use through paired recommender systems with contrasting objectives. In *AAMAS*, pages 3673–3675. ACM, 2026.

[C4] T. Giorgi, P. Olivieri, K. Jiang, L. Toni, and M. Papini. Impact of connectivity on laplacian representations in reinforcement learning. *ICML (to appear)*, 2026.

[C5] A. Montenegro, M. Mussi, M. Papini, and A. M. Metelli. Convergence analysis of policy gradient methods with dynamic stochasticity. In *ICML*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net, 2025.

[C6] G. Neu, M. Papini, and L. Schwartz. Optimistic information directed sampling. In *COLT*, volume 247 of *Proceedings of Machine Learning Research*, pages 3970–4006. PMLR, 2024.

[C7] A. Montenegro, M. Mussi, M. Papini, and A. M. Metelli. Last-iterate global convergence of policy gradients for constrained reinforcement learning. In *NeurIPS*, 2024.

[C8] A. Montenegro, M. Mussi, A. M. Metelli, and M. Papini. Learning optimal deterministic policies with stochastic policy gradients. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 36160–36211. PMLR, 21–27 Jul 2024.

[C9] D. Maran, A. M. Metelli, M. Papini, and M. Restelli. Projection by convolution: Optimal sample complexity for reinforcement learning in continuous-space mdps. In *COLT*, volume 247 of *Proceedings of Machine Learning Research*, pages 3743–3774. PMLR, 2024.

[C10] D. Maran, A. M. Metelli, M. Papini, and M. Restelli. No-regret reinforcement learning in smooth MDPs. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 34760–34789. PMLR, 21–27 Jul 2024.

[C11] D. Maran, A. M. Metelli, M. Papini, and M. Restelli. Local linearity: the key for no-regret reinforcement learning in continuous mdps. In *NeurIPS*, 2024.

[C12] G. Gabbianelli, G. Neu, M. Papini, and N. Okolo. Offline primal-dual reinforcement learning for linear mdps. In *AISTATS*, volume 238 of *Proceedings of Machine Learning Research*, pages 3169–3177. PMLR, 2024.

[C13] G. Gabbianelli, G. Neu, and M. Papini. Importance-weighted offline learning done right. In *ALT*, volume 237 of *Proceedings of Machine Learning Research*, pages 614–634. PMLR, 2024.

[C14] F. Bacchiocchi, F. E. Stradi, M. Papini, A. M. Metelli, and N. Gatti. Online learning with off-policy feedback in adversarial mdps. In K. Larson, editor, *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, pages 3697–3705. International Joint Conferences on Artificial Intelligence Organization, 8 2024. Main Track.

[C15] G. Gabbianelli, G. Neu, and M. Papini. Online learning with off-policy feedback. In *ALT*, volume 201 of *Proceedings of Machine Learning Research*, pages 620–641. PMLR, 2023.

[C16] A. Tirinzoni, M. Papini, A. Touati, A. Lazaric, and M. Pirotta. Scalable representation learning in linear contextual bandits with constant regret guarantees. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:2307–2319, 2022.

[C17] G. Neu, I. Olkhovskaia, M. Papini, and L. Schwartz. Lifting the information ratio: An information-theoretic analysis of thompson sampling for contextual bandits. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:9486–9498, 2022.

[C18] M. Papini, A. Tirinzoni, M. Restelli, A. Lazaric, and M. Pirotta. Leveraging good representations in linear contextual bandits. In *ICML*, volume 139 of *Proceedings of Machine Learning Research*, pages 8371–8380. PMLR, 2021.

[C19] M. Papini, A. Tirinzoni, A. Pacchiano, M. Restelli, A. Lazaric, and M. Pirotta. Reinforcement learning in linear mdps: Constant regret and representation selection. In *NeurIPS*, pages 16371–16383, 2021.

[C20] A. M. Metelli, M. Papini, P. D'Oro, and M. Restelli. Policy optimization as online learning with mediator feedback. In *AAAI*, pages 8958–8966. AAAI Press, 2021.

[C21] M. Papini, A. Battistello, and M. Restelli. Balancing learning speed and stability in policy gradient via adaptive exploration. In *AISTATS*, volume 108 of *Proceedings of Machine Learning Research*, pages 1188–1199. PMLR, 2020.

[C22] P. D'Oro, A. M. Metelli, A. Tirinzoni, M. Papini, and M. Restelli. Gradient-aware model-based policy search. In *AAAI*, pages 3801–3808. AAAI Press, 2020.

[C23] L. Bisi, L. Sabbioni, E. Vittori, M. Papini, and M. Restelli. Risk-averse trust region optimization for reward-volatility reduction. In *IJCAI*, pages 4583–4589. ijcai.org, 2020.

[C24] M. Papini, A. M. Metelli, L. Lupo, and M. Restelli. Optimistic policy optimization via multiple importance sampling. In *ICML*, volume 97 of *Proceedings of Machine Learning Research*, pages 4989–4999. PMLR, 2019.

[C25] M. Beraha, A. M. Metelli, M. Papini, A. Tirinzoni, and M. Restelli. Feature selection via mutual information: New theoretical insights. In *IJCNN*, pages 1–9. IEEE, 2019.

[C26] M. Papini, D. Binaghi, G. Canonaco, M. Pirotta, and M. Restelli. Stochastic variance-reduced policy gradient. In *ICML*, volume 80 of *Proceedings of Machine Learning Research*, pages 4023–4032. PMLR, 2018.

[C27] A. M. Metelli, M. Papini, F. Faccio, and M. Restelli. Policy optimization via importance sampling. In *NeurIPS*, pages 5447–5459, 2018.

[C28] M. Papini, M. Pirotta, and M. Restelli. Adaptive batch size for safe policy gradients. In *NeurIPS*, pages 3591–3600, 2017.

Workshop Papers

[W1] L. Cesani, M. Papini, M. Restelli, How Log-Barrier Helps Exploration in Policy Optimization. EWRL 19 (European Workshop on Reinforcement Learning), Lille, France, 2026.

[W2] L. Cesani, D. Tenedini, M. Papini, M. Restelli, On Natural Policy Compression. EWRL 19 (European Workshop on Reinforcement Learning), Lille, France, 2026.

[W3] L. Cesani, D. Tenedini, M. Papini, M. Restelli, On Natural Policy Compression. Reinforcement Learning Beyond Rewards (RLBrew) workshop at RLC 2026, Montreal, Canada, 2026.

[W4] T. Giorgio, P. Olivieri, K. Jiang, L. Toni, M. Papini. Impact of Connectivity on Laplacian Representations in Reinforcement Learning. CoLorAI workshop at ICML 2026, Seoul, South Korea, 2026.

[W5] A. Menta, F. Maifredi, M. Papini. The Cost of Blind Confidence: Opponent Modeling Under Imperfect Information. NExT-Game workshop at ICML 2026, Seoul, South Korea, 2026.

[W6] D. Salaorni, V. De Paola, S. Delpero, G. Dispoto, P. Bonetti, A. Russo, G. Calcagno, F. Trovò, M. Papini, A. M. Metelli, M. Mussi, M. Restelli. Gym4ReaL: A Benchmark Suite for Evaluating Reinforcement Learning in Realistic Domains. EWRL 18 (European Workshop on Reinforcement Learning), Tübingen, Germany, 2025.

[W7] P. Olivieri, F. Lasca, A. Gianola, M. Papini. Do it for HER: First-order Logic Reward Specification in Reinforcement Learning. RLC Workshop on Programmatic Reinforcement Learning, Edmonton, Canada, 2025.

[W8] L. Bolis, S. Livella, S. Patania, D. Ognibene, M. Papini, K. Morita. Balancing Benefits and Risks: RL Approaches for Addiction-Aware Social Media Recommenders. RLDM (Multidisciplinary Conference on Reinforcement Learning and Decision Making), Dublin, Ireland, 2025.

[W9] M. Molaei, G. Paczolay, M. Papini, A. M. Metelli, M. Restelli. Actor-Critic with Active Importance Sampling. RLDM (Multidisciplinary Conference on Reinforcement Learning and Decision Making), Dublin, Ireland, 2025.

[W10] G. Tedeschi, M. Papini, A. M. Metelli, M. Restelli. Policy Gradient Methods with Adaptive Policy Spaces. EWRL 17 (European Workshop on Reinforcement Learning), Toulouse, France, 2024.

[W11] G. Neu, M. Papini, L. Schwartz. Optimistic Information Directed Sampling. FoRLaC (Foundations of Reinforcement Learning and Control) workshop at ICML, Vienna, Austria, 2024.

[W12] G. Tedeschi, M. Papini, A. M. Metelli, M. Restelli. Policy Gradient Methods with Adaptive Policy Spaces. ARLET (Aligning Reinforcement Learning Experimentalists and Theorists) workshop at ICML, Vienna, Austria, 2024.

[W13] F. Bacchiocchi, F. E. Stradi, M. Papini, A. M. Metelli, N. Gatti. Online Adversarial MDPs with Off-Policy Feedback and Known Transitions. 16th European Workshop on Reinforcement Learning, Brussels, Belgium, 2023

[W14] G. Gabbianelli, G. Neu, N. Okolo, M. Papini. Offline Primal-Dual Reinforcement Learning for Linear MDPs. 16th European Workshop on Reinforcement Learning, Brussels, Belgium, 2023

[W15] G. Neu, J. Olkhovskaya, M. Papini and L. Schwartz. Lifting the Information Ratio: An Information-Theoretic Analysis of Thompson Sampling for Contextual Bandits. 15th European Workshop on Reinforcement Learning, Milan, Italy, 2022

[W16] A. Tirinzoni, M. Papini, A. Touati, A. Lazaric, and M. Pirotta. Scalable Representation Learning in Linear Contextual Bandits with Constant Regret Guarantees. 15th European Workshop on Reinforcement Learning, Milan, Italy, 2022

[W17] G. Gabbianelli, M. Papini, G. Neu. Online Learning with Off-Policy Feedback. ICML-2022 workshop on Complex Feedback in Online Learning, Baltimore, USA, 2022

[W18] A. Gianola, M. Montali, and M. Papini. Automated Reasoning for Reinforcement Learning Agents in Structured Environments. OVERLAY workshop on fOrmal VERification, Logic, Automata and sYnthesis, Padova, Italy, 2021

[W19] M. Papini, A. Tirinzoni, A. Pacchiano, M. Restelli, A. Lazaric, and M. Pirotta. Reinforcement Learning in Linear MDPs: Constant Regret and Representation Selection. ICML Workshop on Reinforcement Learning Theory, virtual, 2021

[W20] M. Papini, A. Battistello, and M. Restelli. Safe Exploration in Gaussian Policy Gradient. NeurIPS-2019 Workshop on Safety and Robustness in Decision Making, Vancouver, Canada, 2019

[W21] M. Papini, A. Battistello, and M. Restelli. Safely Exploring Policy Gradient. 14th European Workshop on Reinforcement Learning, Lille, France, 2018

Papers In Preparation

[P1] With L. Cesani, M. Restelli. Variance-Reduced Policy Gradient is Nearly Minimax-Optimal . Preprint: `https://arxiv.org/abs/2610.13692`

[P2] With N. Cesa-Bianchi. m-Set Adversarial Bandits with Winner Feedback. Preprint: `https://arxiv.org/abs/2610.10128`

[P3] With A. Montenegro, R. Venturelli, M. Mussi, A. M. Metelli. Reusing Past Samples in Proximal Policy Optimization: When and How Does It Help? . Preprint: `https://arxiv.org/abs/2610.01399`

[P4] With D. Salaorni, V. De Paola, S. Delpero, G. Dispoto, P. Bonetti, A. Russo, G. Calcagno, F. Trovò, M. Papini, A. M. Metelli, M. Mussi, M. Restelli. Gym4ReaL: Towards Real-World Reference Environments for Reinforcement Learning. Preprint: `https://arxiv.org/abs/2507.00257`

[P5] With A. Montenegro, L. Cesani, M. Mussi, A. M. Metelli. Learning Deterministic Policies with Policy Gradients in Constrained Markov Decision Processes. Preprint: `https://arxiv.org/abs/2506.05953`

Service Activities

At the Computer Science Department of the University of Milan

2026-present **Exit Guidance Committee member**

2025-present **First-aider**

Attendance to Summer Schools and Exchange Programs

Jul 24–Aug 2 2018 **CIFAR Deep Learning and Reinforcement Learning Summer School**, Toronto, Canada, presented 1 poster

7-14 Oct 2017 **ACAI Summer School on Reinforcement Learning**, Nieuwpoort, Belgium

Autumn 2016 **Erasmus Programme**, KTH Royal Institute of Technology, Stockholm, Sweden

Languages

Italian, *native speaker*

English, *fluent (Cambridge FCE, Grade A — C1 CEFR level)*

Spanish, *intermediate (completed course of level B1 at UPF Barcelona)*